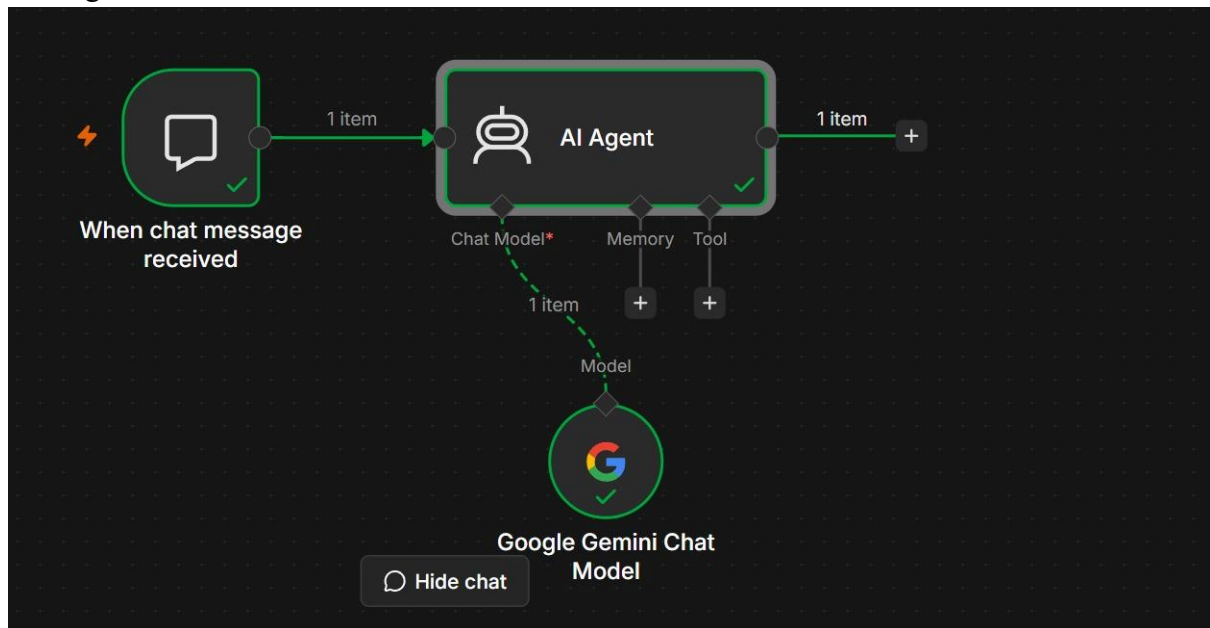


PROJECT-28

AI Regex Generator



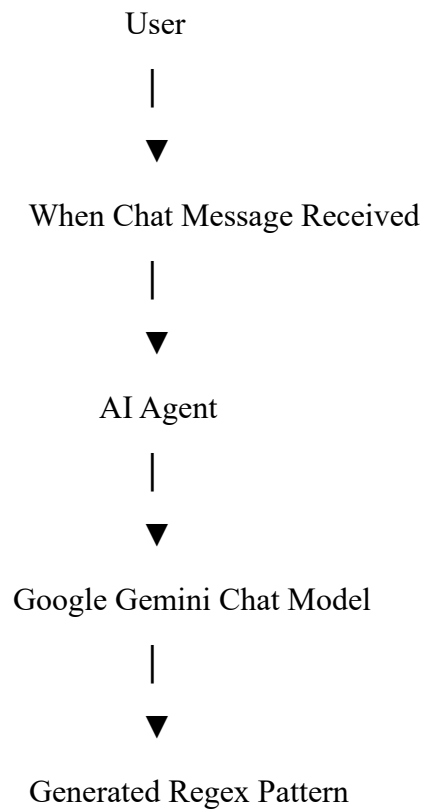
Project Objective

The objective of this project is to build an **AI-powered Regex Generator** that converts plain English descriptions into accurate Regular Expressions (Regex). The AI Agent understands the user's pattern-matching requirements and generates optimized regex patterns along with explanations and sample matches.

The AI Agent can generate:

- Email validation regex
- Phone number regex
- URL matching regex
- Password validation regex
- Date format regex
- ZIP/Postal code regex
- Credit card regex
- Username validation regex
- IP address regex
- Custom regex patterns

Workflow Architecture



Workflow Components

Step	Node	Purpose
1	When Chat Message Received	Receives the user's regex requirement
2	AI Agent	Interprets the request and prepares the regex generation prompt
3	Google Gemini Chat Model	Generates the regex pattern with explanation

Prerequisites

Before building this project, ensure you have:

- n8n (Cloud or Self-hosted)
 - Google AI Studio API Key
 - Google Gemini Chat Model configured
 - Internet connection
-

Step 1 – Create Workflow

Open **n8n**

Click

New Workflow

Rename it

AI Regex Generator

Step 2 – Add Chat Trigger

Add

When Chat Message Received

Purpose

Receives the user's regex generation request.

Example Input

Generate a regular expression to validate email addresses.

Output

```
{  
  "chatInput": "Generate a regular expression to validate email addresses."  
}
```

Step 3 – Add AI Agent

Connect

When Chat Message Received

|



AI Agent

Configure

Source for Prompt

Connected Chat Trigger Node

Prompt

```
{{ $json.chatInput }}
```

Step 4 – Configure the System Prompt

Paste the following into the **System Message**.

You are an AI Regex Generator.

Your task is to convert natural language descriptions into accurate and optimized Regular Expressions (Regex).

Instructions:

- Understand the user's pattern-matching requirement.
 - Generate a valid regular expression.
 - Ensure the regex is efficient and readable.
 - Briefly explain how the regex works.
 - Provide at least three example strings that match the pattern.
 - Provide at least two example strings that do not match the pattern.
 - Mention any assumptions made if the requirement is ambiguous.
 - Return only the regex, explanation, matching examples, and non-matching examples.
-

Step 5 – Add Google Gemini Chat Model

Click

+ **Chat Model**

Choose

Google Gemini Chat Model

Configuration

Setting	Value
Provider	Google Gemini
Model	Gemini 2.5 Flash (or another available Gemini model)

Setting	Value
---------	-------

Authentication	Google AI Studio API Key
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Connect it to the **AI Agent**.

Final Workflow

When Chat Message Received

|



AI Agent

|



Google Gemini Chat Model

Workflow Execution

Step 1

User enters

Generate a regex to validate Indian mobile phone numbers.

↓

Step 2

The **When Chat Message Received** node receives the user's request.

↓

Step 3

The **AI Agent** analyzes the requirement and prepares the regex generation prompt.

↓

Step 4

The prompt is sent to the **Google Gemini Chat Model**.

↓

Step 5

Gemini generates the regex pattern, explains it, and provides matching and non-matching examples.



Step 6

The completed regex solution is returned to the user.

Sample Inputs and Outputs

Example 1

Input

Generate a regex for validating email addresses.

Output

Regex

```
^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$
```

Explanation

Matches standard email addresses with a valid username, domain, and extension.

Matching Examples

- john@example.com
- user123@gmail.com
- admin@test.org

Non-Matching Examples

- john@
- @gmail.com

Example 2

Input

Generate a regex for Indian mobile numbers.

Output

Regex

```
^[6-9]\d{9}$
```

Explanation

Matches a 10-digit Indian mobile number beginning with digits 6 to 9.

Matching Examples

- 9876543210
- 8123456789
- 6999999999

Non-Matching Examples

- 5123456789
 - 987654321
-

Example 3

Input

Generate a regex for URLs.

Output

Regex

```
^https?:\:\/\/[^\s/$.?\#].[\^\s]*$
```

Explanation

Matches valid HTTP and HTTPS URLs.

Matching Examples

- <https://google.com>
- <http://example.org>
- <https://openai.com>

Non-Matching Examples

- <ftp://example.com>
 - <google.com>
-

Example 4

Input

Generate a regex for passwords containing at least one uppercase letter, one lowercase letter, one digit, one special character, and a minimum length of 8 characters.

Output

Regex

`^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&]).{8,}$`

Explanation

Ensures the password includes uppercase and lowercase letters, digits, special characters, and is at least eight characters long.

Matching Examples

- Pass@123
- MyPwd#2024
- Strong\$99

Non-Matching Examples

- password
- PASS1234

Workflow Diagram

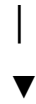
User



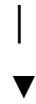
When Chat Message Received



AI Agent



Google Gemini Chat Model



Generated Regex Pattern

Error Handling

Error: No Prompt Specified

Cause

The AI Agent did not receive the user's request.

Solution

Verify that the **Prompt** field contains:

```
{{ $json.chatInput }}
```

Error: Authentication Failed

Cause

Invalid Google Gemini API Key.

Solution

Reconnect or update your Google AI Studio credentials.

Error: Too Many Requests

Cause

Gemini API rate limit exceeded.

Solution

- Wait a few minutes.
 - Check your API quota.
 - Retry the request.
-

Learning Outcomes

After completing this project, you will be able to:

- Build an AI-powered Regex Generator in n8n.
- Configure an AI Agent with Google Gemini.
- Design prompts for regex generation.
- Generate regular expressions from natural language.

- Build conversational AI workflows for developer productivity.
-

Real-World Applications

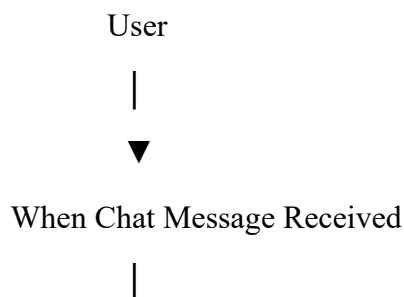
- Input validation
 - Form validation
 - Log file analysis
 - Data extraction
 - Text processing
 - Software development
 - Web application development
 - Automation scripts
 - Data cleaning
 - Developer productivity tools
-

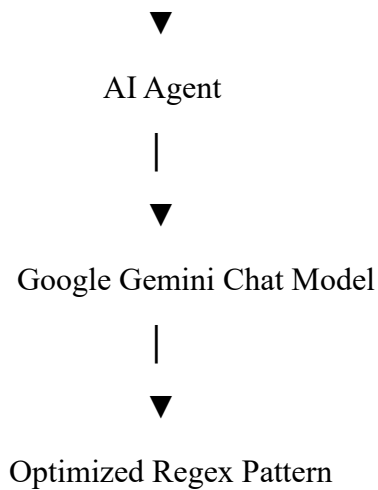
Practice Challenge

Enhance this project by adding:

1. A programming language selector (JavaScript, Python, Java, PHP, C#, etc.).
 2. A regex testing feature with sample input.
 3. A regex explanation mode for beginners.
 4. A Memory node to remember the user's preferred programming language.
 5. A Google Sheets node to save generated regex patterns.
 6. An HTTP Request node to test the generated regex against sample data automatically.
-

Final Workflow





Expected Result

The **AI Regex Generator** accepts a natural language description of a pattern-matching requirement, uses the **Google Gemini Chat Model** to generate an accurate and optimized regular expression, and returns the regex pattern along with a clear explanation, matching examples, and non-matching examples. This project teaches prompt engineering, AI-assisted code generation, and conversational workflow design in n8n, making it an excellent foundation for building intelligent developer tools and automation workflows.